



Automatic Robotic Arm Based on Programmed Instructions

ROBOHARVEST

2 APRIL 2026

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Mentor

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PROBLEM STATEMENT

The Challenge in Agricultural Sorting

Manual Process Limitations

Manual sorting and grading of agricultural products is **time-consuming**, labour-intensive, and often inaccurate – leading to inconsistent quality and significant post-harvest losses.

The Smart Solution

A smart robotic arm system can **automatically detect and sort** fruits and vegetables based on quality, size, and colour – delivering speed, precision, and reliability at scale.



Why Sorting & Grading is Necessary



Prevents Spoilage

Sorting removes damaged or rotten produce early, preventing spoilage from spreading to healthy items and reducing overall waste.



Boosts Profit

Graded produce commands higher market prices. Consistent quality builds buyer trust and improves revenue for farmers and distributors.



Improves Appearance

Well-sorted produce looks more appealing on shelves, attracting more buyers and improving the overall perception of product quality.



Extends Product Life

Proper grading ensures only fresh, undamaged produce enters storage, extending shelf life and reducing losses during transport.

A close-up of a black and silver robotic arm with a gripper, positioned over several black plastic crates filled with ripe red and yellow tomatoes. The scene is set in a greenhouse with bright, diffused light from above. The background is slightly blurred, showing more of the tomato plants and the structure of the greenhouse.

BENEFITS

How a Smart Robotic Arm Helps



Continuous Operation

Unlike human workers, the robotic arm operates **24/7 without fatigue**, ensuring uninterrupted sorting throughput regardless of shift hours.



Labour & Cost Saving

Automation significantly reduces the need for manual labour, **cutting operational costs** while maintaining consistent high-speed performance.



High Accuracy

Sensor-driven detection ensures **precise grading** based on colour, size, and quality – far more reliable than the human eye under fatigue.



Easy to Build & Maintain

Built using affordable, widely available components like Arduino and standard sensor modules, making it **cost-effective to assemble and service**.

HARDWARE

Components Required

Core Hardware

- Arduino Microcontroller
- 4 DOF Robotic Arm Kit (with servo motors)
- Breadboard & Jumper Wires

Power Section

- 12V 2A DC Adapter
- LM2596 Buck Converter
- DC Jack Connector

Sensors

TCS3200

Colour Detection

IR Sensor

Object Detection

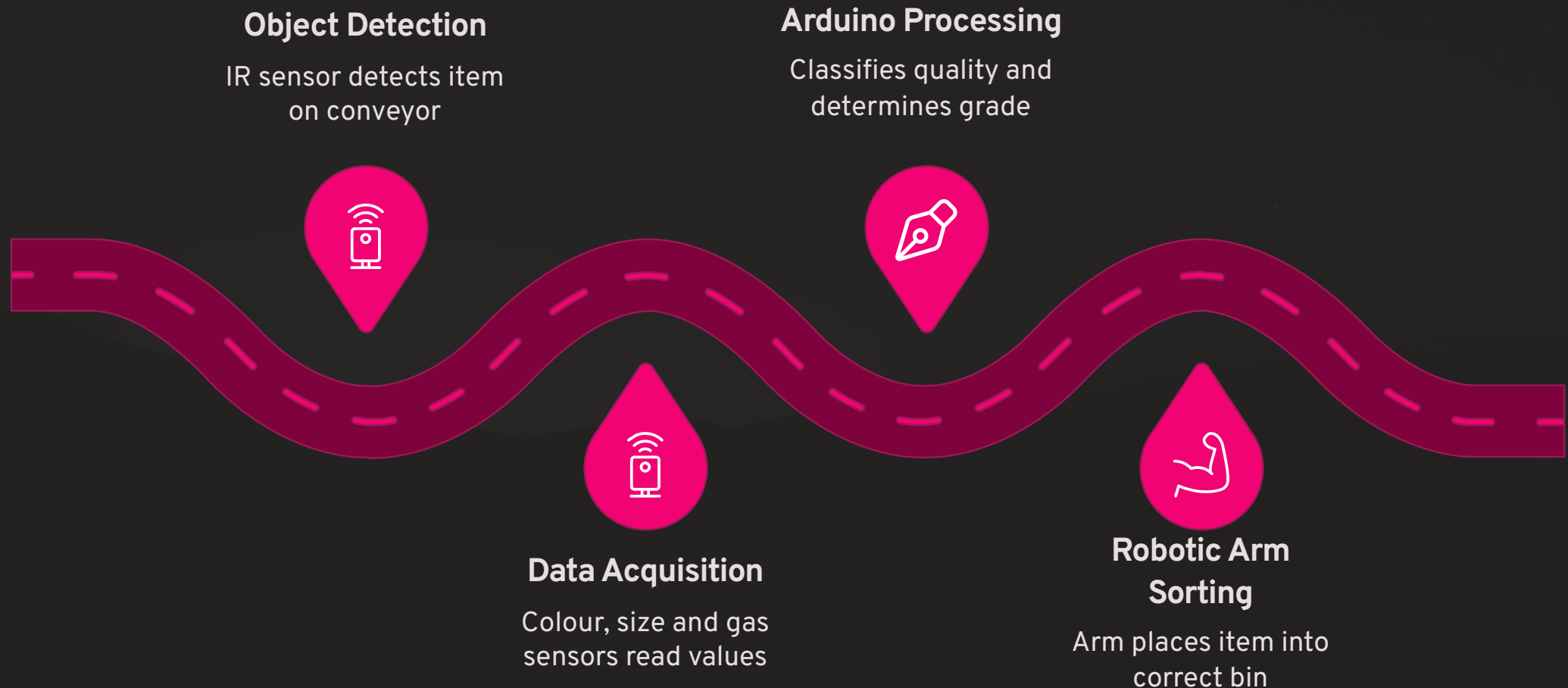
HC-SR04

Size / Distance

MQ-2/135

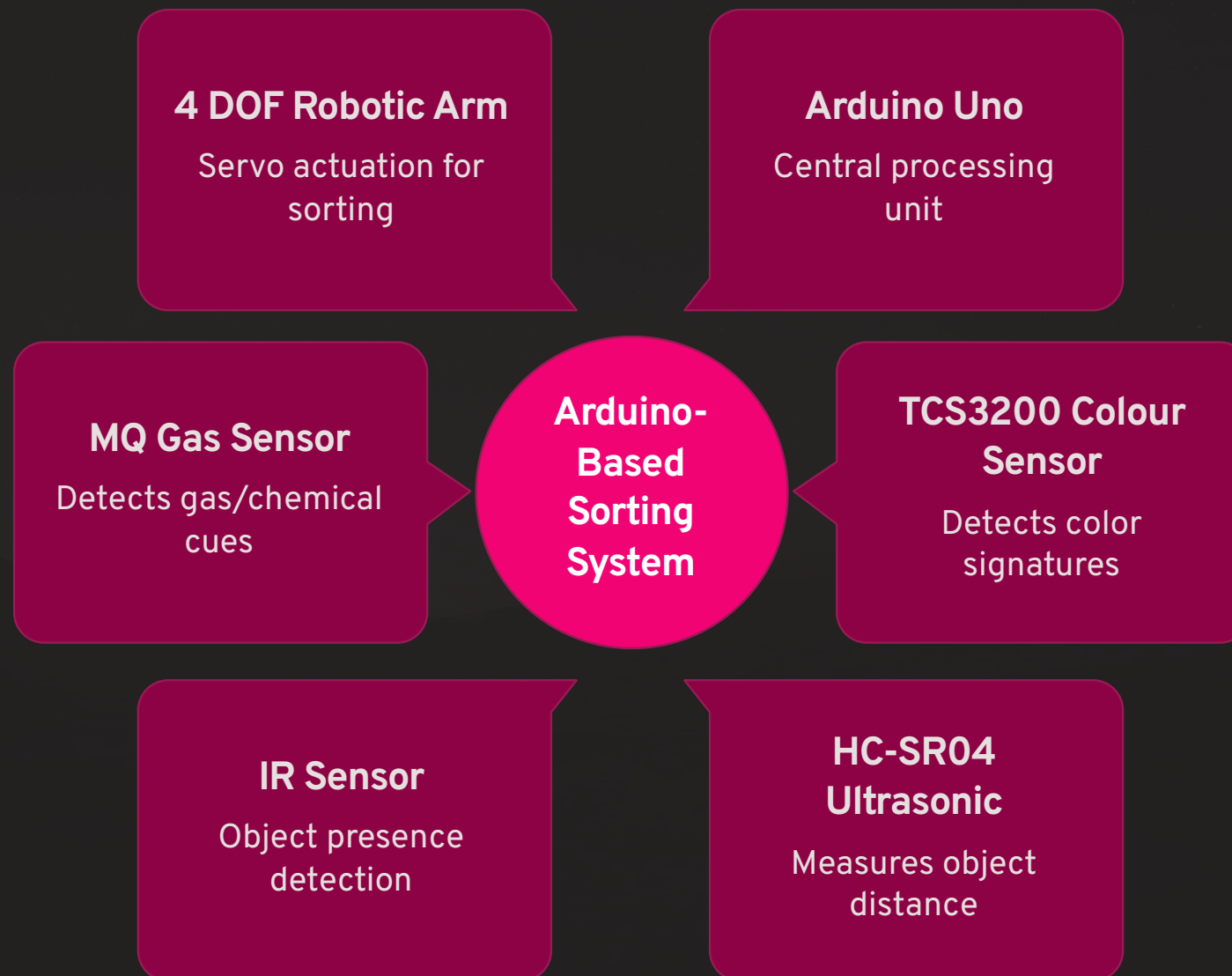
Gas / Ripeness

How the System Works



The Arduino acts as the central brain – reading all sensor inputs in real time, classifying each item, and sending precise servo commands to the robotic arm to complete the sorting action.

System Architecture Diagram



The block diagram above illustrates the complete signal flow – from multi-sensor data acquisition through Arduino processing to the final actuation of the robotic arm for bin-based sorting.

CONCLUSION

Conclusion

What We Built

A cost-effective, Arduino-based robotic arm that automates agricultural sorting using multi-sensor fusion – reducing manual effort and improving grading accuracy.

Impact & Future Scope

This system addresses real-world post-harvest challenges. Future enhancements can include **machine learning-based vision**, IoT connectivity, and scalability to larger conveyor lines.





Thank You!

We extend our sincere gratitude to our mentor **Dr. Naveen Kumar G.N** for his invaluable guidance, and to the judges and faculty for their time and support.

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